

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Year : 2019

Programme: BE

Full Marks: 100

Course: Microprocessor and Assembly Language
Programming

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Compare Microprocessor, Microcomputer and Microcontroller. Which is better for high speed operation? Give reason. 7
- b) Define Instruction cycle. Explain flags, registers and control signals available in 8085 microprocessors. 8
2. a) What is addressing mode? Explain the addressing modes of 8086 microprocessor giving appropriate examples. 7
- b) Draw the timing diagram for the 8085 instruction MVI M,14H. 8
3. a) Write an Assembly language program to multiply two 8 bit numbers stored in the memory address D050H and E050H and if the result is less than 80H, save it to F050H else store it in FFFFH. 7
- b) What is the role of assemblers in ALP? Explain one-pass and two-pass assemblers in brief. What is macro assembler? 8
4. a) Write an assembly language program for 8086 to compare two strings and display "Strings are same" for same strings value otherwise display "Strings are Different" in Dos Screen. 7
- b) What is memory interfacing? Interface 4K x 8 EPROM, 8K x 8 RAM with starting address of 1000H. 8
5. a) What is interrupt? What are the source of interrupts? Explain Maskable/Non-maskable, vectored and non-vectored interrupt. 8
- b) Describe the process to transmit and receive the serial data in 8251 USART. 7
6. a) Describe 8254 PIT with diagram. Write down the instructions to generate 4 KHz square waveform for 8254 PIT 8
- b) What is DMA? Describe how it works with suitable illustration and block diagram. 7

7. Write short notes on: (**Any two**)

2×5

- a) Interrupt Vector Table
- b) MACRO vs PROCEDURE
- c) Memory Classification.